ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | 979-8-3503-6874-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICASSP49660.2025.10889360

Time-varying EEG Signal Reconstruction Based on Local Graph Signal Smoothness

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***Abstract*—Electroencephalography (EEG) signals record the electrical activity of the brain and have significant applications in neuroscience and medicine. However, accurately reconstructing EEG signals has been a challenge due to potential signal missing and noise interference during signal acquisition. The paper exploits the underlying structure of EEG signals and presents an efficient method for reconstructing EEG signals based on local graph signal smoothness (LGS-based). Firstly, we introduce the concept of local graph signal smoothness according to distinct functional areas of the cerebral cortex. Then, considering the graph that does not properly represent similar relationships between signals will have a negative impact on the reconstruction performance, we propose a joint graph learning and EEG signal reconstruction optimization method. Since it is not jointly convex, we utilize the alternating direction method of multipliers (ADMM) to solve it. In experiments, it is shown that the proposed LGS-based method outperforms benchmark methods in EEG signal reconstruction. Additionally, the proposed method achieves a higher signal-to-noise ratio (SNR) as well as a smaller normalized mean square error (RMSE).**

***Index Terms*—EEG signals reconstruction, local graph signal smoothness, graph learning, alternating direction method of multipliers (ADMM).**

I. Introduction

The human brain is a complex organ, and there have been many experiments and studies to understand it. The electroencephalogram (EEG) is considered a helpful tool for better understanding this complex system. EEG utilizes multiple electrodes that cover the surface of the brain, allowing for the capture of brain activity with high temporal resolution [1]–[3]. However, in practical applications, the quality of EEG signal recording is often affected by additional observational noise (such as electromyographic noise and artifacts) and signal missing. This can lead to incomplete or noisy signals, ultimately affecting the quality of the signal. Particularly in medical contexts, poor EEG signal quality can hinder accurate diagnosis of a patient's condition, thus affecting the effectiveness and efficiency of medical tasks. In order to enhance the quality of EEG signals, it is particularly important to effectively reconstruct EEG signals from partial and noisy observations. This can help restore the integrity of the signals to the greatest extent possible. Most existing EEG signal reconstruction methods rely on traditional signal processing tools, such as the maximum likelihood (ML-based) [4], the robust multichannel reconstruction (RCLR) [5], the alternating direction method of multipliers (ADMM-based) [6], the robust wavelet transform (WT) [7], the myriad filter [8], and independent component analysis [9]. While these methods are effective, they do not fully exploit the underlying structure of EEG signals, including potential intra-channel and inter-channel interdependencies. To address this limitation, graph-based methods have been gaining an increasing interest in the biomedical signal processing community. Specifically, graph-based methods represent a signal ensemble by means of graphs and leverage the extensive analytical techniques developed in the emerging field of graph signal processing (GSP) [10], [11]. The fundamental advantage of the GSP framework lies in its ability to leverage both local and global interconnections among signals, as captured by vertices within an irregular graph structure. This becomes particularly crucial in the context of reconstructing graph signals, where accounting for the correlation between vertices can significantly enhance the quality of signal reconstruction. EEG signals can naturally be represented as a graph, where each electrode corresponds to a vertex, and edges define the connections between these vertices. For the application of GSP in EEG signal reconstruction, a number of studies have shown promising results, namely, $l_{p,\epsilon}$-regularized graph filter (lpe-graph-based) [13], Laplacian denoising algorithm [14] and electrical brain source reconstruction [15]. Despite the advantages of GSP, its effective application relies heavily on the use of an appropriate graph that can accurately represent the inherent relationship between the signals. While it is easy to define graph vertices as recording electrodes, there is no straightforward definition of graph edges and edge weights. Therefore, effective graph learning is one of the challenges in

This work is supported by the National Natural Science Foundation of China under Grant 62471147. *(Corresponding author: Xue-Qin Jiang.)*

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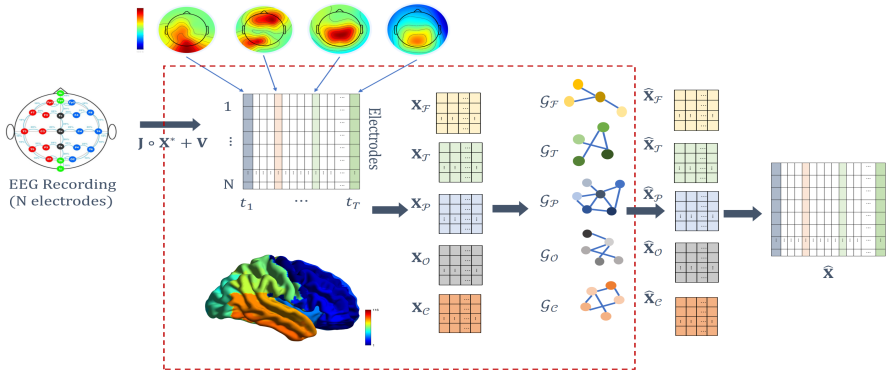

Fig. 1. The schematic overview of the proposed LSG-based method.

current EEG signal reconstruction.

Inspired by the good results of the application of GSP and graph learning in brain mapping, we propose the reconstruction method for time-varying EEG signals based on local graph signal smoothness. First, we introduce the concept of local graph signal smoothness, which is based on the segmentation of different functional regions of the cerebral cortex, so that the focus of graph signal processing is placed on the localized smoothing process. Second, we propose a joint graph learning and EEG signal reconstruction optimization method. Then, given that the proposed optimization method is not jointly convex, we resolve it using the ADMM algorithm. A schematic overview of the paper is shown in Fig. 1. The incomplete EEG signals obtained through electrodes can be expressed as $\mathbf{J} \circ \mathbf{X}^* + \mathbf{V}$, where $\mathbf{J}$, $\mathbf{X}^*$ and $\mathbf{V}$ represent the sampling matrix, source signal and noise, respectively. Based on the functional division of the brain, the signals are divided into frontal $\mathbf{X}_{\mathcal{F}}$, parietal $\mathbf{X}_{\mathcal{P}}$, temporal $\mathbf{X}_{\mathcal{T}}$, occipital $\mathbf{X}_{\mathcal{O}}$ and center $\mathbf{X}_{\mathcal{C}}$. The local graphs $\mathcal{G}_{\mathcal{F}}$, $\mathcal{G}_{\mathcal{P}}$, $\mathcal{G}_{\mathcal{T}}$, $\mathcal{G}_{\mathcal{O}}$ and $\mathcal{G}_{\mathcal{C}}$ and the reconstructed signals $\hat{\mathbf{X}}_{\mathcal{F}}$, $\hat{\mathbf{X}}_{\mathcal{P}}$, $\hat{\mathbf{X}}_{\mathcal{T}}$, $\hat{\mathbf{X}}_{\mathcal{O}}$ and $\hat{\mathbf{X}}_{\mathcal{C}}$ are both obtained through the proposed method in this paper. Finally, by integrating the locally reconstructed signals, the complete signal $\hat{\mathbf{X}}$ is reconstructed.

This paper is organized as follows. In Section II, some preliminaries are provided. Section III introduces the concept of local graph signal smoothness and formulates the problem of reconstructing the EEG signal as an optimization problem jointing graph learning and graph signal reconstruction. We solve the proposed joint optimization problem by ADMM. The simulation results are provided in Section IV. Section V concludes the paper.

For clarity of presentation, the list of notations used in this paper is provided in TABLE I.

TABLE I
LIST OF NOTATIONS

Notations	Description
$\mathbf{A}/\mathbf{a}$	a matrix / a vector
$\mathrm{vec}(\mathbf{A})$	vectorized form of the matrix $\mathbf{A}$
$\mathrm{diag}(\mathbf{A})$	the diagonal elements of the matrix $\mathbf{A}$
$\mathrm{upper}(\mathbf{A})$	the upper triangular part of the matrix $\mathbf{A}$
$\mathbf{A}^\top/\mathrm{Tr}(\mathbf{A})$	the transpose / trace of the matrix $\mathbf{A}$
$\lVert\cdot\rVert_F/\lVert\cdot\rVert_*$	the Frobenius norm / nuclear norm
$\circ$	Hadamard product

II. PRELIMINARIES

An undirected, connected, and weighted graph is denoted as $\mathcal{G}(\mathcal{V}, \mathcal{E}, \mathbf{W})$, where $\mathcal{V} = \{1, \cdots, N\}$ represents the vertex set of cardinality N and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ the set of edges. $\mathbf{W} \in \mathbb{R}^{N\times N}$ denote the adjacency matrix of edge weights. The symmetric and non-negative coefficients $W_{ij} = W_{ji}$ indicate the strength of the connection between electrode i and electrode j. In the absence of connection $[i.e., (i,j) \nsubseteq \mathcal{E}]$ one has $W_{ij} = 0$. The adjacency matrix $\mathbf{W}$ encodes the brain functional connectivity. Beyond the adjacency matrix $\mathbf{W}$, results in spectral graph theory often motivate choosing the combinatorial graph Laplacian $\mathbf{L} := \mathrm{diag}(\mathbf{W}\mathbf{1}) - \mathbf{W}$. A time-varying EEG signal can be expressed by a matrix $\mathbf{X}^* = [\mathbf{x}_1^*, \mathbf{x}_2^*, \cdots, \mathbf{x}_T^*] \in \mathbb{R}^{N\times T}$, where $\mathbf{x}_t^*$ denotes the signal at time instant t, and each row of $\mathbf{X}^*$ can be viewed as a time series on corresponding channel. In the context of EEG signal processing, graph signals correspond to the EEG signals and graph represents the brain functional connectivity. Accordingly vertices stand for the EEG electrodes. To reduce the impact of high-frequency noise and transient fluctuations in the original time-varying signal on the smoothness assumption, we introduce differential temporal signals $\mathbf{x}_t^* - \mathbf{x}_{t-1}^*$, $t \in \{1, \cdots, T\}$. Then, the smoothness of time-varying graph signal is defined as

$$f(\mathbf{X}^*) = \mathrm{Tr}[(\mathbf{X}^*\mathbf{D})^\top \mathbf{L}\mathbf{X}^*\mathbf{D}], \tag{1}$$

where $\mathbf{D}$ is the temporal difference operator, denoted as

$$\mathbf{D} = \begin{bmatrix} -1 & & & \\ 1 & -1 & & \\ & 1 & \ddots & \\ & & \ddots & -1 \\ & & & 1 \end{bmatrix} \in \mathbb{R}^{T\times(T-1)}. \tag{2}$$

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III. EEG SIGNAL RECONSTRUCTION BASED ON LOCAL GRAPH SMOOTHNESS

In this section, we present a graph-based reconstruction method of EEG signals. First, we proposed local graph signal smoothness according to the characteristic property of EEG signals. Then, a new formulation for joint graph signal reconstruction and local graph learning problem is proposed which can be solved using ADMM algorithm.

A. Local Graph Signal Smoothness

This subsection introduces the concept of local graph learning, which involves using local graphs to extract features and relationships between different regions in the signal. It is well-established that the cerebral cortex shows unique responses in different functional areas when exposed to various external stimuli. Consequently, the EEG signals within the same functional area demonstrate local smoothness. Based on this observation, we divided the EEG signal into 5 areas corresponding to different functional areas of the cerebral cortex, including the frontal lobe, temporal lobe, parietal lobe, occipital lobe, and central lobe. Therefore, the EEG signal can be expressed as $\mathbf{X}^* = \{\mathbf{X}_1^*, \cdots, \mathbf{X}_k^*\}$ with $k = \{1, \cdots, K\}$ represents signals in different regions. Each regions of signals exhibits a distinct structure, categorized based on the functional areas of the cerebral cortex and the characteristics of electrode sampling points. Similar to [12], given a matrix of graph signal, $\mathbf{X}_k^*$, the regions-specific Laplacian matrix under signal smoothness prior can be learned by solving the following optimization problem

$$\min_{\mathbf{L}_k \in \mathcal{L}} \mathrm{Tr}(\mathbf{Z}_k^\top \mathbf{L}_k \mathbf{Z}_k) + \alpha \|\mathbf{L}_k\|_F \quad s.t. \quad \mathrm{Tr}(\mathbf{L}_k) = N_k, \tag{3}$$

where $\mathcal{L} = \{\mathbb{L}_k | \mathbf{L}_k = \mathbf{L}_k^\top, L_{ij} = L_{ji} \geq 0 (\forall i \neq j), \mathbf{L}_k \mathbf{1} = \mathbf{0}\}$ is the set of Laplacian matrices and $\mathbf{Z}_k = \mathbf{X}_k^* \mathbf{D} \in \mathbb{R}^{N_k \times (T-1)}$ and N_k represents the number of vertices in the k-th local graph. $\mathbf{L}_k \in \mathbb{R}^{N_k \times N_k}$ represents the Laplacian matrix of the k-th local graph and $\alpha > 0$ is the regularization parameter.

B. Reconstruction Method Based on Local Graph Signal Smoothness

In this subsection, we focus on the case where there is K computing center to collect the sampled signal $\mathbf{X}_k$ of all the vertices, and the local graph Laplacian matrix $\mathbf{L}_k$ is unknown. A reconstruction method for time-varying EEG signals based on the local graph signal smoothness (LGS-based) is proposed. In real world applications, measurements of EEG signals are usually contaminated by Gaussian noise or missing values. Therefore, we defined the sampling of the time-varying EEG signal $\mathbf{X}^*$ as

$$\mathbf{X} = \mathbf{J} \circ \mathbf{X}^* + \mathbf{V}, \tag{4}$$

where $\mathbf{V}$ is the additive Guassian white noise, and $\mathbf{J} \in \{0,1\}^{N \times T}$ is the sampling operator defined as

$$J_{u,t} = \begin{cases} 1 & u \in \mathcal{S}_t \\ 0 & u \notin \mathcal{S}_t \end{cases}, \tag{5}$$

TABLE II
LIST OF VECTORIZED FORM

Vector	Description	Vector	Description
$\mathbf{u}_{\mathbf{L}_k}$	upper($\mathbf{L}_k$)	$\mathbf{d}_{\mathbf{L}_k}$	diag($\mathbf{L}_k$)
$\hat{\mathbf{x}}_k$	upper($\hat{\mathbf{X}}_k \hat{\mathbf{X}}_k^\top$)	$\mathbf{d}_{\hat{\mathbf{X}}_k}$	diag($\hat{\mathbf{X}}_k \hat{\mathbf{X}}_k^\top$)
$\hat{\mathbf{z}}_k$	upper($\hat{\mathbf{Z}}_k \hat{\mathbf{Z}}_k^\top$)	$\mathbf{d}_{\hat{\mathbf{Z}}_k}$	diag($\hat{\mathbf{Z}}_k \hat{\mathbf{Z}}_k^\top$)
$\mathbf{d}_{\mathbf{J}_k}$	diag($\mathbf{J}_k$)	$\hat{\mathbf{g}}_k$	vec($\hat{\mathbf{X}}_k$)
$\mathbf{g}_k$	vec($\mathbf{X}_k$)	$\boldsymbol{\theta}$	vec($\boldsymbol{\Theta}$)
$\mathbf{p}_k$	vec($\mathbf{P}_k$)		

where $\mathcal{S}_t$ denotes that set of vertice sampled at time instant t. Then, the problem of EEG signal reconstruction and local graph learning can be written as the following optimization problem

$$\begin{aligned} &\min_{\hat{\mathbf{X}}_k, \mathbf{L}_k \in \mathcal{L}} \mathrm{Tr}(\hat{\mathbf{Z}}_k^\top \mathbf{L}_k \hat{\mathbf{Z}}_k) + \alpha \|\mathbf{L}_k\|_F + \beta \|\hat{\mathbf{X}}_k\|_* \\ &+ \frac{\psi}{2} \|\mathbf{J} \circ \mathbf{X}_k - \hat{\mathbf{X}}_k\|_F \quad s.t. \quad \mathrm{Tr}(\mathbf{L}_k) = N_k, \end{aligned} \tag{6}$$

where $\beta > 0$ and $\psi > 0$ are regularization parameters. The term $\|\hat{\mathbf{X}}_k\|_*$ models $\hat{\mathbf{X}}_k$ to be low-rank based on the assumption that $\hat{\mathbf{X}}_k$ includes redundant information as each graph signal is generated from the same graph topology. This problem is not jointly convex in $\mathbf{L}_k$ and $\hat{\mathbf{X}}_k$. Therefore, problem (6) are solved with alternating direction method of multipliers (ADMM). First, we fix $\hat{\mathbf{X}}_k$ and solve the problem for $\mathbf{L}_k$, then fixing $\mathbf{L}_k$ we solve with respect to $\hat{\mathbf{X}}_k$. Since we are learning a set of symmetric matrices $\mathbf{L}_k$ with $k = 1, \cdots, K$, the problem can be reformulated in vector form, focusing on learning the upper triangle and diagonal components of $\mathbf{L}_k$. We list the vertorized forms of the matrices that need to be used in TABLE II.

1) $\mathbf{L}_k$-subproblem: Assuming $\hat{\mathbf{X}}_k$ to be fixed, we need to solve the following problem for the $\mathbf{L}_k$-subproblem

$$\mathbf{L}_k^* = \arg \min_{\mathbf{L}_k \in \mathcal{L}} m(\mathbf{L}_k) \quad s.t. \quad \mathrm{Tr}(\mathbf{L}_k) = N_k, \tag{7}$$

where $m(\mathbf{L}_k) = \mathrm{Tr}(\hat{\mathbf{Z}}_k^\top \mathbf{L}_k \hat{\mathbf{Z}}_k) + \alpha \|\mathbf{L}_k\|_2^2$.

In this paper, we set $\|\mathbf{L}_k\|_2^2 = \frac{1}{2}\left(\|\mathbf{u}_{\mathbf{L}_k}\|_2^2 + \|\mathbf{d}_{\mathbf{L}_k}\|_2^2\right)$ with the constraint $\mathbf{1}^\top \mathbf{d}_{\mathbf{L}_k} = N_k$. Then, the augmented Lagrangian of problem (7) is

$$\begin{aligned} \mathcal{L}(\mathbf{u}_{\mathbf{L}_k}, \mathbf{d}_{\mathbf{L}_k}, \boldsymbol{\lambda}, \rho_1) &= 2\hat{\mathbf{z}}_k \mathbf{u}_{\mathbf{L}_k} + \mathbf{d}_{\hat{\mathbf{Z}}_k}^\top \mathbf{d}_{\mathbf{L}_k} + \frac{\alpha_1}{2} \mathbf{u}_{\mathbf{L}_k}^\top \mathbf{u}_{\mathbf{L}_k} \\ &+ \frac{\alpha_2}{2} \mathbf{d}_{\mathbf{L}_k}^\top \mathbf{d}_{\mathbf{L}_k} + \boldsymbol{\lambda}^\top (\mathbf{1}^\top \mathbf{d}_{\mathbf{L}_k} - N_k) + \phi^\top (\mathbf{F}\mathbf{u}_{\mathbf{L}_k} + \mathbf{d}_{\mathbf{L}_k}) \\ &+ \frac{\rho_1}{2} \|\mathbf{1}^\top \mathbf{d}_{\mathbf{L}_k} - N_k\|_2^2 + \frac{\rho_1}{2} \|\mathbf{F}\mathbf{u}_{\mathbf{L}_k} + \mathbf{d}_{\mathbf{L}_k}\|_2^2, \end{aligned} \tag{8}$$

where $\mathbf{F}\mathbf{u}_{\mathbf{L}_k} = \mathbf{L}_k \mathbf{1} - \mathbf{d}_{\mathbf{L}_k}$ and $\phi > 0$, $\rho_1 > 0$ are penalty factors. The problem (8) is convex and the steps of ADMM are as follows

$$\mathbf{u}_{\mathbf{L}_k}^{c+1} = -\left(\alpha_1 \mathbf{I} + \rho_1 \mathbf{F}^\top \mathbf{F}\right)^{-1} \mathbf{Y}, \tag{9}$$

$$\mathbf{d}_{\mathbf{L}_k}^{c+1} = \left[(\alpha_1 + \rho_1)\mathbf{I} + \rho_1 \mathbf{1}\mathbf{1}^\top\right]^{-1} \mathbf{H}, \tag{10}$$

$$\boldsymbol{\lambda}^{c+1} = \boldsymbol{\lambda}^c + \rho_1 (\mathbf{F}\mathbf{u}_{\mathbf{L}_k}^{c+1} + \mathbf{d}_{\mathbf{L}_k}^{c+1}), \tag{11}$$

$$\phi^{c+1} = \phi^c + \rho_1 (\mathbf{1}^\top \mathbf{d}_{\mathbf{L}_k}^{c+1} - N_k), \tag{12}$$

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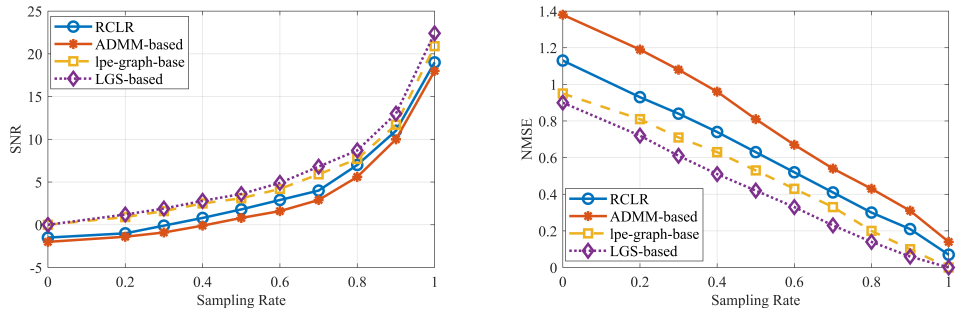

Fig. 2. Performance results of the original EEG signal reconstruction (in terms of SNR and NMSE) versus the sampling rate.

where $\mathbf{Y} = 2\hat{\mathbf{z}}_k + \boldsymbol{\lambda}^c\mathbf{F} + \rho_1\mathbf{F}^\top\mathbf{d}^c_{\mathbf{L}_k}$ and $\mathbf{H} = (\rho_1 N_k - \phi)\mathbf{1} - \mathbf{d}_{\hat{\mathbf{Z}}_k} - \boldsymbol{\lambda}^c - \rho_1\mathbf{F}\mathbf{u}^{c+1}_{\mathbf{L}_k}$.

2) $\hat{\mathbf{X}}$-subproblem: Let $\mathbf{L}_k$ be fixed, the objective function in (6) would be only a function of $\hat{\mathbf{X}}$. Then, the optimization problem can be written as

$$\hat{\mathbf{X}}^*_k = \arg\min_{\hat{\mathbf{X}}_k} h(\hat{\mathbf{X}}^*_k), \tag{13}$$

where $h(\hat{\mathbf{X}}^*_k) = \mathrm{Tr}(\hat{\mathbf{Z}}^\top_k\mathbf{L}_k\hat{\mathbf{Z}}_k) + \beta\|\hat{\mathbf{X}}_k\|_* + \frac{\psi}{2}\|\mathbf{J}\circ\mathbf{X}_k - \hat{\mathbf{X}}_k\|_F$. By introducing the auxiliary variable $\mathbf{P}_k$, the problem (13) is transformed into the following problem with constraints

$$\min_{\hat{\mathbf{X}}_k} \mathrm{Tr}(\hat{\mathbf{X}}^\top_k\mathbf{D}^T_k\mathbf{L}_k\mathbf{D}_h\hat{\mathbf{Z}}_k) + \beta\|\hat{\mathbf{X}}_k\|_* + \frac{\psi}{2}\|\mathbf{J}_k\circ\hat{\mathbf{X}}_k - \mathbf{X}_k\|^2_F \quad s.t. \quad \hat{\mathbf{X}}_k - \mathbf{P}_k = \mathbf{0}. \tag{14}$$

The corresponding augmented Lagrangian is

$$\mathcal{L}(\hat{\mathbf{g}}_k, \mathbf{p}, \boldsymbol{\theta}) = \hat{\mathbf{g}}^\top_k\left[(\mathbf{D}_k\mathbf{D}^\top_k)\otimes\mathbf{L}_k\right]\hat{\mathbf{g}}_k + \beta\|\hat{\mathbf{g}}_k\|_* + \frac{\psi}{2}\|\mathbf{d}_{\mathbf{J}_k}(\hat{\mathbf{g}}_k - \mathbf{g}_k)\|^2_2 + \boldsymbol{\theta}^\top(\hat{\mathbf{g}}_k - \mathbf{p}_k) + \frac{\rho_2}{2}\|\hat{\mathbf{g}}_k - \mathbf{p}_k\|^2_2, \tag{15}$$

where ρ_2 is penalty factor. The steps of ADMM are as follows

$$\hat{\mathbf{g}}^{c+1}_k = \mathbf{Q}^{-1}\left(\boldsymbol{\theta}^c - \psi\mathbf{d}_{\mathbf{J}_k}\mathbf{g}_k - \rho_2\mathbf{p}^c + \beta\mathbf{I}\right), \tag{16}$$

$$\mathbf{p}^{c+1}_k = \hat{\mathbf{g}}^{c+1}_k + \frac{1}{\rho_2}\boldsymbol{\theta}^c, \tag{17}$$

$$\boldsymbol{\theta}^{c+1} = \boldsymbol{\theta}^c + \rho_2(\hat{\mathbf{g}}^{c+1}_k - \mathbf{p}^{c+1}_k), \tag{18}$$

where $\mathbf{Q} = (\mathbf{D}_k\mathbf{D}^\top_k)\otimes\mathbf{L}_k + \psi\mathbf{d}_{\mathbf{J}_k} + \rho_2\mathbf{1}$. In accordance with the aforementioned ADMM solution method, once the reconstructed local signal $\hat{\mathbf{X}}^*_k$ is obtained, it is integrated to derive the final reconstructed EEG signal $\hat{\mathbf{X}}^* = \{\hat{\mathbf{X}}^*_1, \cdots, \hat{\mathbf{X}}^*_k\}$.

IV. Experimentals

In this section, we present the simulation results of our proposed LGS-based method for graph signal reconstruction on EEG signals. To this end, we apply our method to a widely used and publicly available EEG dataset called MODMA dataset [16]. We denote the data of every 100 consecutive timesteps of the EEG data as $\mathbf{X}^* \in \mathbb{R}^{N\times T}$ for simulation. The dataset includes data from 53 testers, with each tester's data forming a graph with 128 vertices. Each vertex is linked to a discrete sequence of 100 timesteps. We use SNR and NMSE criteria to measure the signal reconstruction performance. If we denote the original data matrix with $\mathbf{X}^*$, its reconstructed version by $\hat{\mathbf{X}}$ and their ith columns by $\mathbf{x}^*_i$ and $\hat{\mathbf{x}}_i$, respectively, then we have SNR $= 10\log_{10}\left(\frac{\|\mathbf{X}^*\|^2_2}{\|\mathbf{X}^* - \hat{\mathbf{X}}\|^2_2}\right)$ and NMSE $= \frac{1}{T}\sum_{t=1}^{T}\frac{\|\mathbf{x}^*_t - \hat{\mathbf{x}}_t\|^2_2}{\|\mathbf{x}^*_t\|^2_2}$. To provide more robustness, we average the results over 50 different random sampling realizations. The parameters of our method are chosen as $\alpha_1 = \alpha_2 = 0.04$, $\beta = 0.5$, $\psi = 0.1$, $\rho_1 = 0.5$, $\rho_2 = 0.5$ and the sampling strategies used in all of the following simulations are random sampling. In order to evaluate the reconstruction performance of our proposed LGS-based method, we compare our LGS-based method to several benchmark methods, which also include graph-based modeling. These include RCLR [5], ADMM-based [6] and lpe-graph-based [13]. Fig. 2 depicts the performance of the proposed LGS-based method compared to the benchmark methods for reconstruction of the dataset, at different values of the sampling rate. The proposed method is shown to outperform the other methods for reconstruction of the missing data. This advantage stems from the incorporation of local graph signal smoothness, which makes the reconstruction process more aligned with the actual data distribution. Additionally, the joint optimization of graph learning and signal reconstruction effectively extracts and utilizes the structural information of graph, thereby significantly enhancing the performance of the reconstruction.

V. Conclusion

In this paper, we proposed the reconstruction method for time-varying EEG signals based on local graph smoothness. Due to the different areas of the cerebral cortex being responsible for processing different types of information and performing different functions, we proposed local graph learning based on the smoothness of signals in the same area. Subsequently, based on the smoothness of local graph signals, we further proposed a joint graph learning and EEG signal reconstruction method called LSG-based method. By comparing the two performance indexes of SNR and NMSE, the experiments proved that the proposed reconstruction method is more effective than the benchmark methods.

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